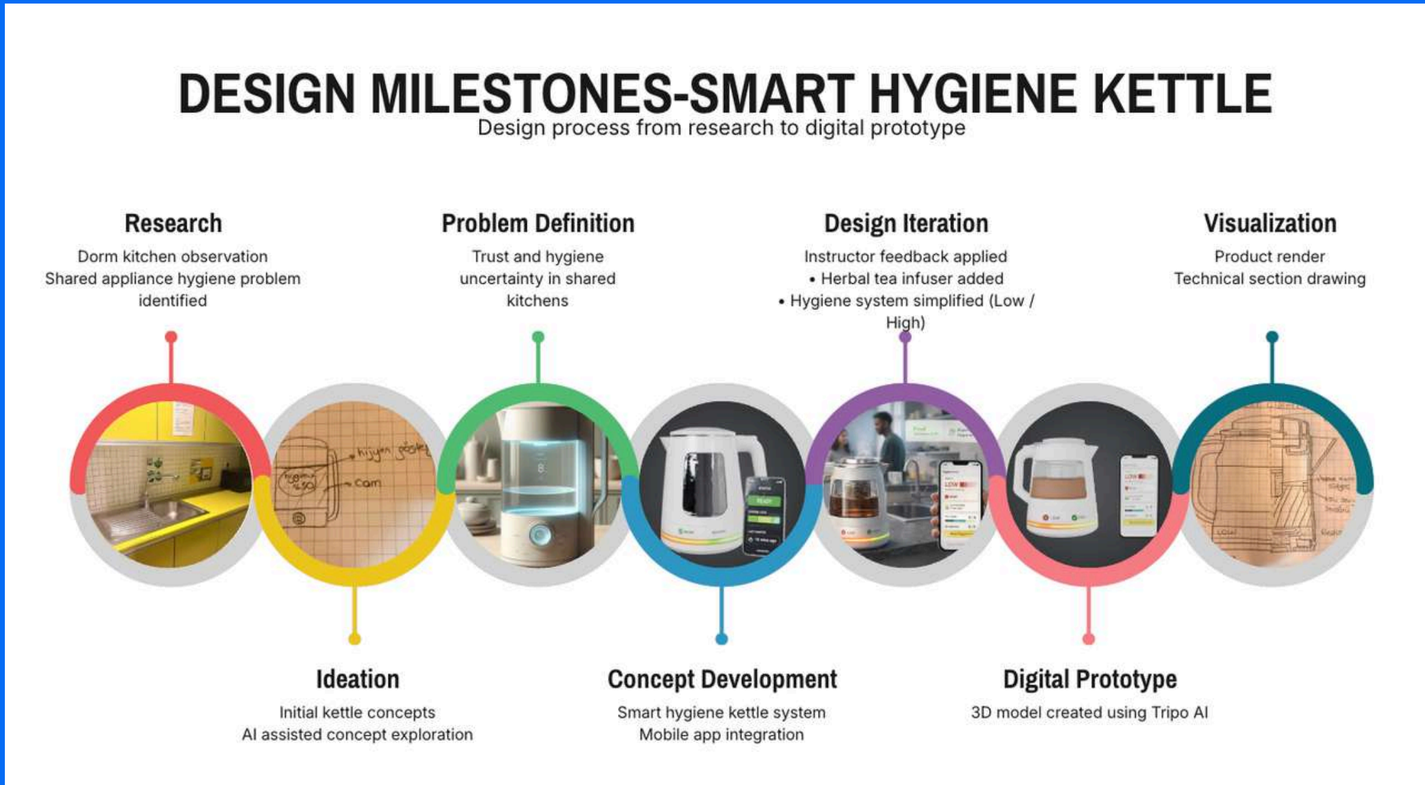
SMART HYGIENE KETTLE

DESIGN MILESTONES DOCUMENTATION

DESIGN PROCESS OVERVIEW

This project explores hygiene trust issues in shared dormitory kitchens. The design process began with identifying behavioral problems in collective appliance use. The concept evolved through ideation, AI-supported visualization, and instructor feedback. The final outcome is a smart kettle system designed to improve transparency and hygiene awareness in shared environments.



KEYWORDS

- Shared kitchen environments
- Hygiene transparency
- Trust in collective use
- Smart appliance design
- Steam sanitation system
- User behavior feedback
- IoT connectivity
- Dormitory living

These keywords helped frame the design problem and guided the concept development process.

SELECTED AI PROMPTS

PROMPT 1

Design a smart electric kettle for shared dorm kitchens that includes a hygiene indicator system and steam sanitation cycle.

PROMPT 2

Create an exploded component view of a smart kettle including heating element, thermostat, sensors and connectivity module.

PROMPT 3

Visualize a smart kettle with an integrated herbal tea infuser and mobile app hygiene monitoring system.

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AI ROLES IN THE DESIGN PROCESS

Clarifier

AI helped clarify the internal structure and component organization of the kettle.

Re-framer

AI helped reframe the project from a basic kettle design into a hygiene-oriented shared appliance.

Expander

AI expanded the exploration by generating alternative visualizations and component layouts.

Critique

AI assisted in evaluating design logic and identifying possible improvements.

AI RESPONSE SUMMARY

AI tools were used mainly for concept visualization, product rendering, and component breakdown illustrations. These outputs helped visualize the internal structure of the kettle, including the heating system, sanitation mechanism, and electronic components. AI also supported the creation of environmental product renders to show the kettle in a dorm kitchen context.

REFLECTION & DESIGN DECISIONS

Reflection

AI tools helped accelerate the visualization and concept exploration process. They allowed quick testing of different product architectures and component layouts. However, the final design decisions were mainly guided by usability considerations and instructor feedback.

Design Decisions

Accepted

- Steam sanitation system
- Herbal tea infuser integrated into the lid
- Mobile hygiene monitoring system

Rejected

- Fully transparent kettle body
- Complex multi-level hygiene indicators

Transformed

- Hygiene indicator simplified from three levels to two levels (Low / High)